

Table 4: NeuroGate – Linear Probing with Frozen Backbone

Dataset	CBraMod LP bAcc	NeuroGate LP bAcc	Δ
HMC	0.6149 ± 0.0256	0.6922 ± 0.0172	+0.0773
ISRUC-S1	0.5962 ± 0.0579	0.6934 ± 0.0156	+0.0972
MDD	0.8907 ± 0.0477	0.9062 ± 0.1101	+0.0154
Physionet-MI	0.2719 ± 0.0176	0.2875 ± 0.0083	+0.0156

Note: NeuroGate shows the largest gains on sleep staging tasks (HMC, ISRUC-S1), suggesting that explicit neural priors can supplement information not easily read out by a linear head from frozen CBraMod embeddings.